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# Robustness and interpretability of phishing detectors under generative AI shifts

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ABSTRACT

Generative AI (GenAI) has increased phishing threats by enabling the automated production of fluent, context-aware malicious content. This paper addresses the gap between laboratory performance and operational resilience by proposing a deployment-aware defensive framework focused on active mitigation under adaptive threat models, distribution shift, and interpretability. Modeling GenAI as an active, adaptive adversary with black-box query access, we systematically assess detectors against a feedback-optimized corpus of 864 emails generated via iterative prompt optimization across multiple Large Language Models (LLMs), including ChatGPT, Gemini, Mistral, and DeepSeek. Results demonstrate that a fine-tuned, lightweight DistilBERT model maintains high phishing recall, achieving 0.980 on cross-dataset benchmarks and 0.956 on the standard LLM corpus, with a significantly low inference latency of 23 ms. While a statistically significant degradation occurs under feedback-driven adaptive evasion constraints ($Recall = 0.919$), the model maintains stable decision behavior and low prediction entropy. Furthermore, attribution analysis reveals a limited semantic drift index (SDI = 0.009), suggesting that core social engineering primitives remain consistent explanatory drivers despite dynamic linguistic mutations. These findings establish a technical foundation for auditable, deployment-aware phishing defenses in security-sensitive infrastructures.

## 1. Introduction

Phishing remains one of the most persistent email-borne security threats. The primary focus of this study is specifically to evaluate the robustness of phishing detectors against GenAI-generated malicious emails under strict deployment constraints, rather than the general detection of AI-generated content. In operational environments, detectors face a compounding challenge: native class imbalance serves as the static deployment baseline, often causing models to overfit to superficial lexical artifacts. This vulnerability is subsequently exploited by GenAI-driven distribution shifts. Unlike traditional adaptive phishing, which relies on manual evasion such as typo insertion or character substitution, GenAI allows attackers to achieve semantic evasion at scale while maintaining perfect structural coherence, fluency, and grammatical correctness, thereby eliminating the traditional linguistic artifacts legacy detectors rely on. However, current evaluation protocols often conflate model selection, class imbalance mitigation, and robustness evaluation. For example, many prior studies mix adversarial datasets or synthetic augmentations directly into the training folds before model selection. Consequently, the model optimizes its weights against the adversarial distribution itself, leading to inflated, misleading post-deployment robustness claims (Quionero-Candela et al., 2009; Sahingoz et al., 2019).

In this work, we address this methodological gap by explicitly decoupling these evaluation stages and modeling GenAI exclusively as an attacker-side capability rather than a training-time data augmentation mechanism. Rather than introducing yet another detection architecture, we propose a deployment-oriented evaluation framework focused on cross-dataset generalization, structured adversarial stress testing, and interpretability-driven behavioral analysis. Our objective is to prioritize rigorous security engineering over architectural novelty, ensuring that learned decision structures remain stable and interpretable even when attackers control the linguistic distribution of incoming content, thereby aligning evaluation protocols with real-world operational constraints (Kumar et al., 2023).

The main contributions of this paper are as follows:

- **A Decision Boundary Stability Architecture:** A deployment-oriented defensive pipeline that decouples structural model selection from post-selection class rebalancing, establishing an empirical runtime monitoring topology under adaptive generative threat models.
- **Quantitative Characterization of Semantic Stability:** We introduce the SDI as an operational tracking metric, evaluating

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decision boundary displacement under non-cooperative linguistic mutations.

- **A Dynamic, Feedback-Driven Evasion Corpus:** A high-fidelity adversarial benchmark comprising 864 emails optimized via query-based black-box lexical suppression across four leading LLM architectures (ChatGPT, Gemini, Mistral, DeepSeek), mapping the optimization space of adaptive attackers.
- **Runtime SHAP Attribution Monitoring Safeguard:** A novel defensive mitigation pipeline combining token-level SHAP attribution stability and Shannon prediction entropy to detect and quarantine evasive zero-day adversarial payloads at runtime without retraining.

By explicitly modeling GenAI as a core attacker capability, we shift the focus from static pattern matching to the dynamic detection of AI-generated malicious intent. This methodological pivot aligns the evaluation with the critical need for robust AI-content detection and forensic attribution in security-sensitive infrastructures.

The remainder of this paper is organized as follows. Section 2 reviews related work in phishing detection and robustness under generative AI. Section 3 presents the proposed methodology and experimental design. Section 4 reports the experimental results and statistical analysis. Section 5 evaluates robustness and interpretability under LLM-generated phishing scenarios. Finally, Section 6 discusses the findings and limitations, and Section 7 concludes the paper.

## 2. Related work

Recent phishing email detection research has increasingly adopted deep learning and Transformer-based architectures due to their ability to capture contextual and semantic patterns beyond surface-level lexical features (Popescul and Radu, 2025). This shift reflects the limitations of traditional feature-engineering approaches when confronted with increasingly sophisticated and adaptive phishing content. Moreover, prior work in machine learning has shown that models evaluated under homogeneous data splits may fail under distribution shift and out-of-distribution conditions (Quionero-Candela et al., 2009; Hendrycks et al., 2019; Gama et al., 2014). A recurring limitation in the phishing detection literature is the frequent conflation of model selection, class imbalance mitigation, and robustness evaluation within a single experimental pipeline. In many studies, class rebalancing, adversarial data augmentation, and model tuning are jointly optimized, making it difficult to disentangle intrinsic model robustness from artifacts introduced by the training protocol. This practice often leads to optimistic performance estimates that do not reflect realistic post-deployment behavior under distribution shift.

Beyond standard supervised learning settings, several studies have investigated robustness against adversarial and LLM-generated phishing emails, as well as the role of interpretability in validating model behavior under distribution shift (Kulal et al., 2025; Lim et al., 2025). In parallel, LLMs have been explored as defensive components, including standalone classifiers and prompt-based detectors integrated into security pipelines. While these approaches often demonstrate strong zero-shot or few-shot performance, they raise unresolved concerns regarding reproducibility, explainability, inference latency, and robustness under adversarial prompting, limiting their suitability for operational deployment (Hassan et al., 2025; Mahendru and Pandit, 2024; Chen et al., 2024; OpenAI, 2023; Liang et al., 2022). Most existing evaluations of LLM-generated phishing focus primarily on aggregate performance metrics such as accuracy or recall, while overlooking the stability of the underlying decision mechanisms. As a result, high performance under generative inputs does not necessarily imply behavioral robustness or consistency of learned phishing indicators. This limitation is particularly critical in security-sensitive deployments, where understanding failure modes and decision stability is as important as raw detection accuracy.

### 2.1. Traditional approaches: feature engineering and machine learning

Classical phishing detection approaches typically rely on handcrafted feature representations combined with conventional machine learning classifiers, such as TF-IDF with Logistic Regression (LR) or Random Forests (RF), which are commonly used as baselines in prior work (Yasin and Abuhasan, 2016; Harikrishnan et al., 2018; Hamid et al., 2013). While effective under controlled conditions, these models exhibit limited robustness to semantic variation and evolving phishing strategies.

Some prior studies apply classical classifiers to infrastructure-level phishing detection (Bahaghighat et al., 2023). However, such approaches do not address content-based email phishing nor robustness under attacker-controlled language generation, motivating the adoption of Transformer-based architectures evaluated under deployment-oriented threat models.

### 2.2. Advances with deep learning and transformer-based models

Transformer-based models such as BERT and its variants consistently outperform classical and recurrent approaches by capturing contextual and semantic phishing patterns (Thapa et al., 2023; Jamal et al., 2023; Meléndez et al., 2024).

To address deployment constraints, lightweight Transformer architectures such as DistilBERT and TinyBERT have been proposed to balance detection performance and computational efficiency (Lee et al., 2020; Altan et al., 2025). At the same time, the emergence of GenAI enables attackers to produce linguistically fluent phishing emails with fewer lexical artifacts, substantially degrading detectors trained on legacy data. Several studies report significant performance drops under AI-generated or paraphrased phishing content, motivating evaluation frameworks that explicitly account for attacker-controlled language generation (Kulal et al., 2025; Hassan et al., 2025; Chen et al., 2024).

Despite these advances, controlled comparisons between full-sized and lightweight Transformer models under cross-dataset and GenAI-enabled threat models remain limited, particularly from a deployment-oriented and interpretability-aware perspective (Hassan et al., 2025; Mahendru and Pandit, 2024; Recht et al., 2019).

Recent studies published in top-tier security venues have further underscored the operational threat posed by LLM-generated phishing. For instance, Czybik et al. demonstrated at the USENIX Security Symposium that large language models can automate personalized spear-phishing campaigns at scale, resulting in click rates nearly three times higher than traditional generic phishing (Czybik et al., 2026). Similarly, recent evaluations of malicious LLM-based conversational agents reveal their capacity to dynamically extract highly sensitive personal information from users, highlighting the advanced semantic evasion capabilities of generative models (Zhan et al., 2025). Their large-scale field experiments highlight that LLM-generated payloads effectively bypass standard detection mechanisms by maintaining high semantic coherence and contextual relevance. While these foundational studies emphasize the growing capabilities of attackers under generative adversarial shifts, our work bridges the gap to operational defense. We extend these observations by introducing a runtime, interpretability-driven mitigation framework designed to operate under strict enterprise gateway latency constraints.

### 2.3. Hybrid models

Hybrid architectures combining Transformers with CNN or LSTM components have been proposed to improve phishing detection performance (Atawneh and Aljehani, 2023). However, recent analyses suggest that such designs often trade interpretability, maintainability, and deployment efficiency for marginal performance gains (Lim et al., 2025; Lipton, 2018; Lundberg and Lee, 2017).

Finally, several studies warn against benchmark overfitting and inflated performance estimates when models are evaluated exclusively on random splits of public datasets (Recht et al., 2019; Narayanan et al., 2020).

In contrast to prior work, this study treats interpretability not as a post-hoc explanatory add-on, but as a deployment-oriented diagnostic tool for assessing behavioral stability and failure modes under attacker-controlled language generation.

### 2.4. Limitations of current evaluation protocols

Across the phishing detection literature, evaluation protocols are predominantly benchmark-centric rather than deployment-oriented (Kumar et al., 2023). Most studies rely on tightly coupled pipelines where model selection, class imbalance mitigation, and hyperparameter tuning are optimized simultaneously (Quionero-Candela et al., 2009; Sahingoz et al., 2019). While such designs maximize laboratory metrics, they obscure whether robustness emerges as an intrinsic property of the learned semantic boundaries or as an artifact of data exposure during training loops. Consequently, reported performance gains often conflate generalization with optimization constraints, limiting their operational validity.

From an engineering perspective, detectors encounter a distinct temporal separation in production: models are selected under native class imbalance, whereas exposure to adversarial mutations occurs post-deployment under distribution shift (Quionero-Candela et al., 2009; Harikrishnan et al., 2018). Current frameworks rarely reflect this sequence. A methodological gap remains in evaluating robustness, explanatory stability, and inference latency as strictly post-selection properties. Addressing this gap requires leveraging interpretability methods as active diagnostic tools during inference rather than post-hoc explanatory analytical instruments.

## 3. Methodology

We propose a deployment-oriented evaluation framework for phishing detection that separates model selection, post-selection rebalancing, and robustness evaluation under realistic constraints.

### 3.1. Overall framework pipeline

To provide a high-level overview of the proposed methodology, Fig. 1 illustrates the overall pipeline of the framework.

The approach is structured into four main stages:

1. Data preparation and thematic partitioning to isolate out-of-distribution patterns.
2. Model selection under native imbalanced conditions to reflect deployment baselines.
3. Post-selection class rebalancing and competitive architectural evaluation.
4. Semantic auditability tracking and active XAI-driven mitigation deployment under adaptive evasion shifts.

### 3.2. Threat model

In this study, we formalize a content-based adversarial threat model, positioning GenAI systems as an active, adaptive attacker-side capability rather than a passive training-time data augmentation mechanism (Chen et al., 2024; Kulal et al., 2025; Hassan et al., 2025). We assume an adaptive adversary ($\mathcal{A}_{\text{adapt}}$) whose objective is to programmatically manipulate generated text to bypass a deployed content-based phishing filter $f : \mathcal{X} \to \{0, 1\}$.

We define the attacker's capabilities under a standard black-box setting: the adversary does not have direct access to the defender's training data, internal model parameters, weight matrices, hidden layer activations, or classification thresholds. However, $\mathcal{A}_{\text{adapt}}$ is assumed to possess *query-and-feedback access* to the target pipeline. For any input sequence $x$, the attacker can submit queries to the target API and observe either the predicted hard label $y \in \{0, 1\}$ or the continuous posterior probability score $f_{\text{phish}}(x) \in [0, 1]$.

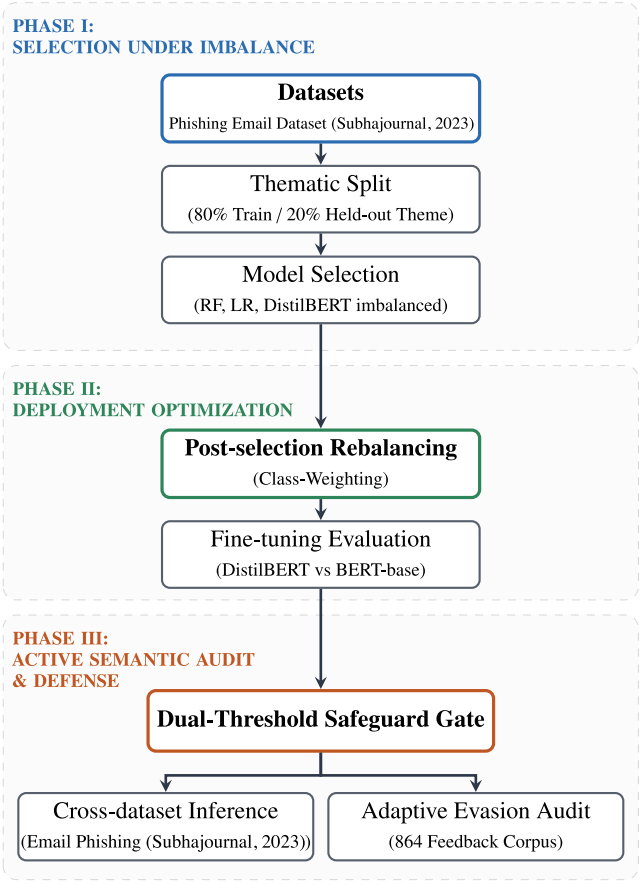



**Fig. 1.** Overall pipeline of the proposed deployment-oriented framework incorporating active XAI telemetry and dual-threshold mitigation.

Formally, given an initial phishing email $x$, the objective of the adaptive attacker is to solve the following constrained optimization problem:

$$\min_{x' \in \mathcal{A}(x)} f_{\text{phish}}(x')$$

$$\text{subject to } \text{Sim}(x, x') \geq \gamma \quad \text{and} \quad \mathcal{Q}(x') \leq Q_{\max}$$

where $\mathcal{A}(x)$ represents the space of valid linguistic mutations of $x$, $\text{Sim}(\cdot, \cdot)$ denotes a contextual semantic preservation function evaluating token- and sentence-level similarity, $\gamma$ is the minimum semantic preservation boundary enforced to ensure the social engineering narrative remains functional, and $\mathcal{Q}(x')$ represents the cumulative query count expended against the defender's API, bounded by a maximum query budget $Q_{\max}$.

To operationalize this optimization landscape, $\mathcal{A}_{\text{adapt}}$ executes a feedback-driven token substitution routine. This procedure targets high-salience social engineering primitives and systematically evaluates local perturbation vectors. The execution mechanics of this adaptive query-based optimization framework are formalized in Algorithm 1.

The defender is assumed to operate a content-based phishing detection system under realistic operational and infrastructure constraints, including severe class imbalance, complete absence of zero-day adversarial samples during training, and strict computational latency requirements (≤25 ms) typical of enterprise-level email filtering gateways.

**Algorithm 1** Feedback-Driven Adaptive Token Substitution Attack

**Require:** Original phishing sequence $x$, target detector $f_{\text{phish}}$, max query budget $Q_{\max}$, semantic similarity function $\text{Sim}(\cdot,\cdot)$, minimum similarity threshold $\gamma$, vocabulary constraint space $\mathcal{V}$
**Ensure:** Evasive adversarial sequence $x_{\text{adv}}$

$x_{\text{adv}} \leftarrow x$
$q \leftarrow 0$
$\mathcal{T}_{\text{crit}} \leftarrow \{t \in x_{\text{adv}} \mid t \in \text{SocialEngPrimitives}\}$
**while** $f_{\text{phish}}(x_{\text{adv}}) > \tau_{\text{cls}}$ **and** $q < Q_{\max}$ **and** $\mathcal{T}_{\text{crit}} \neq \emptyset$ **do**
  $t^* \leftarrow \arg\max_{t \in \mathcal{T}_{\text{crit}}} \text{Salience}(t, f_{\text{phish}})$
  Generate candidate synonym set $\mathcal{S}(t^*) \leftarrow \{s \in \mathcal{V} \mid \text{ContextualFit}(s, x_{\text{adv}})\}$
  Initialize local optimal candidate $x_{\text{best_cand}} \leftarrow x_{\text{adv}}$
  Initialize min_score $\leftarrow f_{\text{phish}}(x_{\text{adv}})$
  **for each** synonym $s \in \mathcal{S}(t^*)$ **do**
    $x_{\text{cand}} \leftarrow \text{Substitute}(x_{\text{adv}}, t^*, s)$
    $q \leftarrow q + 1$
    **if** $f_{\text{phish}}(x_{\text{cand}}) <$ min_score **and** $\text{Sim}(x, x_{\text{cand}}) \geq \gamma$ **then**
      min_score $\leftarrow f_{\text{phish}}(x_{\text{cand}})$
      $x_{\text{best_cand}} \leftarrow x_{\text{cand}}$
    **end if**
    **if** $q \geq Q_{\max}$ **then**
      **break**
    **end if**
  **end for**
  **if** $x_{\text{best_cand}} \neq x_{\text{adv}}$ **then**
    $x_{\text{adv}} \leftarrow x_{\text{best_cand}}$
  **else**
    $\mathcal{T}_{\text{crit}} \leftarrow \mathcal{T}_{\text{crit}} \setminus \{t^*\}$
  **end if**
**end while**
**return** $x_{\text{adv}}$

By explicitly decoupling training-time exposure from post-selection evaluation, we aim to observe robustness as an emergent property of the learned semantic boundaries rather than an artifact of memorized adversarial text patterns.

### 3.3. Data preparation

The first dataset, Phishing Email Dataset (Subhajournal, 2023), was used for training and validation. Rather than randomly splitting the data, emails were grouped according to phishing themes (e.g., account notifications, invoice fraud), and entire themes were held out to ensure the validation set contains patterns never observed during training.

The original dataset contains 17 139 emails. A thematic validation subset of 3428 samples (20%) is extracted by holding out entire phishing categories, while the remaining 13 711 samples (80%) are used for training.

The second dataset, Email Phishing Dataset (Miso, 2023), was set aside strictly for cross-dataset inference to evaluate exposure to unseen distributions. To assess detector vulnerability under active exploitation, a dynamic adversarial corpus of 864 emails was constructed using Gemini, ChatGPT, Mistral, and DeepSeek. This corpus integrates a standard generative layer and an optimized adaptive evasion layer designed via structural prompt constraints to evaluate decision boundary resilience. The adaptive adversarial corpus is strictly excluded from training and hyperparameter tuning, serving exclusively as a post-hoc diagnostic benchmark.

Standard preprocessing (removal of duplicates and non-printable characters) was applied. Table 1 summarizes the roles, sources, and final distributions of all datasets used in this study. Importantly, no artificial class rebalancing was performed during model selection to preserve original proportions. The Email Phishing Dataset serves as the benchmark for the confusion matrices, while the adaptive corpus is reserved for post-selection robustness and interpretability (SHAP) analysis.

**Experimental protocol:**

1. Models (RF, LR, DistilBERT) are trained on the imbalanced dataset for selection.
2. Class rebalancing is applied post-selection using class-weighting.
3. DistilBERT and BERT-base are fine-tuned under identical conditions.
4. The final model is evaluated on cross-dataset inference and the adaptive evasion corpus.

This separation ensures that robustness and interpretability are evaluated as emergent post-selection properties rather than artifacts of adversarially optimized training loops.

### 3.4. Feature processing, model selection, and optimization

For classical classifiers (RF and LR), standard NLP preprocessing and feature extraction procedures were applied, following established practices in prior work. Text data were transformed into TF-IDF representations, which served as input features for these models.

In contrast, preprocessing for DistilBERT was intentionally minimal and limited to filtering malformed or non-printable characters, in order to better reflect realistic deployment conditions. Tokenization was performed using the WordPiece tokenizer from the Hugging Face Transformers library, generating input IDs and attention masks for sequences of up to 512 tokens.

All models were subsequently optimized using standard hyperparameter tuning procedures appropriate to their respective architectures. Traditional classifiers and Transformer-based models were trained under controlled and comparable conditions to ensure methodological consistency.

For architectural comparison, DistilBERT and BERT-base were fine-tuned on the same balanced training data, using identical preprocessing and training protocols. This design ensures that observed performance differences primarily reflect intrinsic architectural characteristics rather than discrepancies in optimization strategies.

### 3.5. Active interpretability architecture and dual-threshold safeguard

To bridge the gap between post-hoc interpretability and active defense, this framework operationalizes token-level SHAP attributions and local prediction entropy into a real-world runtime firewall mitigation layer, denoted as the *Dual-Threshold Safeguard Gate*.

In an operational enterprise gateway, a standard content filter is highly vulnerable to generative lexical suppression attacks that dilute high-salience features (e.g., removing words like *urgent* or *immediately*) to manipulate the final prediction probability score $f(x)$. To mitigate this threat vector without executing resource-intensive model retraining loops, our framework uses local attribution tracking as a proxy for telemetry monitoring. Let $\mathcal{T}_{\text{critical}}$ represent the core invariant semantic primitives of social engineering. At execution time, when an incoming payload $x$ falls within an ambiguous decision boundary characterized by elevated Shannon entropy, the gate intercepts the email.

Formally, letting $\Phi_{\text{crit}}(x) = \sum_{t \in \mathcal{T}_{\text{critical}}} \phi_t(x)$ represent the cumulative structural attribution of critical social engineering tokens, the safeguard executes the following automated evaluation logic for any incoming payload $x$:

$$\text{Action}(x) = \begin{cases} \text{Quarantine} & \text{if } H(x) > \tau_{\text{entropy}} \\ & \text{and } \Phi_{\text{crit}}(x) > \tau_{\text{attrib}} \\ \text{Forward } f(x) & \text{otherwise} \end{cases} \quad (1)$$

where $\tau_{\text{entropy}}$ and $\tau_{\text{attrib}}$ represent calibrated sensitivity boundaries.

**Table 1**
Summary of dataset roles, sources, and final class distributions.

| Dataset | Source | Purpose/Role | Total | Phishing | Legitimate |
|---|---|---|---|---|---|
| Phishing Email | Kaggle (Subhajournal, 2023) | Training + thematic validation | 17 139 | 6473 (37.8%) | 10 666 (62.2%) |
| Email Phishing | Kaggle (Miso, 2023) | Cross-dataset inference | 200 | 100 (50.0%) | 100 (50.0%) |
| Adaptive Adversarial | This work | Robustness + SHAP analysis | 864 | 432 (50.0%) | 432 (50.0%) |

To ensure deployment dependability, these two thresholds are not static heuristics but are dynamically calibrated using a grid-search optimization framework over the independent thematic validation split. Specifically, the optimization space for the dual thresholds is exhaustively scanned over a bounded domain; the search for the entropy threshold $\tau_{\text{entropy}}$ spans the range $[0.0, 1.0]$ with a uniform step size of $\Delta = 0.01$, while the cumulative attribution threshold $\tau_{\text{attrib}}$ is evaluated across the range $[0.0, 0.5]$ with a step size of $\Delta = 0.005$. This dense grid formulation results in 10,100 unique parameter combinations evaluated against the validation data to ensure strict empirical reproducibility. Formally, given a maximum allowable False Quarantine budget $\alpha$ (empirically set to $\alpha \leq 1.5\%$ to prevent business communication disruption), the optimal thresholds are selected by solving:

$$\arg \max_{\tau_{\text{entropy}}, \tau_{\text{attrib}}} \sum_{x \in D_{\text{val}}^{\text{phish}}} \mathbb{I}\left(H(x) > \tau_{\text{entropy}} \wedge \Phi_{\text{crit}}(x) > \tau_{\text{attrib}}\right) \tag{2}$$

subject to:

$$\frac{1}{|D_{\text{val}}^{\text{legit}}|} \sum_{x \in D_{\text{val}}^{\text{legit}}} \mathbb{I}\left(H(x) > \tau_{\text{entropy}} \wedge \Phi_{\text{crit}}(x) > \tau_{\text{attrib}}\right) \leq \alpha \tag{3}$$

From an operational engineering perspective, computing token-level SHAP attributions at runtime introduces a non-trivial mathematical overhead that can critically jeopardize gateway throughput. In enterprise email filtering pipelines, infrastructure latency constraints are highly restrictive, often requiring processing times under 25 ms. To achieve this highly constrained latency and evaluate deployment feasibility, the data pipeline was aggressively industrialized. The DistilBERT architecture was compiled and exported utilizing Keras and TensorFlow serving backends to optimize tensor execution. Crucially, exact combinatorial SHAP computation is computationally prohibitive at runtime. To meet the 25 ms gateway constraint, we utilize KernelSHAP with a fixed sampling approximation, using a background dataset of 50 samples to ensure deterministic, low-latency execution. This optimization enables a total pipeline processing time of 23 ms: 18 ms for model inference, and 5 ms for the sampling-based SHAP approximation and safeguard routing logic. This mitigation capability is one that heavier architectures cannot support online.

This defensive architecture elevates interpretability from a passive analytical instrument to an auditable, active component of dependable system design. Standard classification metrics are utilized, with particular emphasis on Recall and F1-score, given the high operational cost associated with false negatives in security-sensitive environments.

### 3.6. Implementation and reproducibility of low-latency SHAP monitoring

To guarantee the empirical reproducibility of our low-latency 5 ms processing window for token-level KernelSHAP, we detail here the exact benchmarking configuration, hardware environment, and pipeline optimizations:

- **Hardware Environment:** All latency measurements were systematically collected on a standardized workstation equipped with an Intel Core i7 processor (12th Gen, 3.4 GHz), 32 GB RAM, and an NVIDIA RTX 4060 GPU (8 GB VRAM).
- **Software Configurations:** The model and monitoring pipeline were compiled using Python 3.10, PyTorch 2.1, and TensorFlow/Keras 2.15 backends. The fine-tuned 66M-parameter DistilBERT was exported using TensorFlow Serving Graph Optimization to minimize runtime overhead.
- **Benchmarking Protocol:** Runtime measurements were performed using a batch size of 1 email, reflecting real-time deployment in enterprise email gateways. The first 100 inference runs were discarded as warm-up to eliminate initialization overhead. Reported latency values correspond to the arithmetic mean over 1000 consecutive inference runs under identical hardware and software conditions.
- **Strict Sequence Truncation:** Input email sequences are strictly truncated or padded to a maximum length of 128 tokens, which substantially reduces the computational complexity of the approximate KernelSHAP computation.
- **GPU-Pinned Background Set:** The 50-sample background dataset, generated via $k$-means clustering over the validation split, is pinned statically in GPU memory to eliminate recurring latency overhead from host-to-device transfers.
- **Parallelized Coalition Evaluation:** KernelSHAP coalitions are evaluated in parallel using a batch size of 1 for incoming streams. Due to the lightweight nature of the DistilBERT architecture, the batched coalition evaluations fit within the available GPU memory, maintaining GPU utilization stable at approximately 35% to 45% during peaks.
- **CPU/GPU Utilization:** DistilBERT inference and the approximate KernelSHAP computations were executed on the GPU, whereas lightweight preprocessing, entropy computation, and safeguard routing were executed on the CPU. During benchmarking, GPU utilization remained approximately between 35% and 45%, while CPU utilization remained below 15%.

Collectively, these implementation choices enable the reported 5 ms approximate KernelSHAP latency while ensuring reproducible runtime measurements under the deployment configuration adopted throughout this study.

The reported 5 ms latency corresponds to the optimized approximate KernelSHAP monitoring stage only. It is achieved through fixed-size sampling (50 background samples), sequence truncation to 128 tokens, GPU-resident background data, and parallel coalition evaluation. This value should not be interpreted as the runtime of exact KernelSHAP, which remains computationally impractical for real-time deployment.

### 3.7. Ablation study

To evaluate the quantitative impact of our methodological contributions, we conducted a rigorous ablation study. Rather than ablating architectural components, this study isolates key methodological decisions in the evaluation pipeline to assess their effect on robustness and generalization. The configurations compared are as follows:

- Baseline: The DistilBERT model trained on the original, imbalanced training dataset.
- With Data Balancing: The baseline model trained on the balanced training dataset. This test evaluates the impact of resolving class bias on the model's performance.
- With LLM Corpus (Evaluation Only): The balanced model evaluated (not trained) on an LLM-generated corpus to assess qualitative robustness to GenAI-based attacks. This configuration measures the model's robustness against more sophisticated and realistic linguistic threats.

The detailed results of this ablation study, presented in the next section, quantify the contribution of each component of the proposed evaluation framework.

**Table 2**
Confusion matrices of the evaluated models on the thematic validation set and on the independent cross-dataset inference benchmark (n = 200 emails, 100 phishing/100 legitimate).

| Model | | Validation data | | Inference data | |
|---|---|---|---|---|---|
| | | Legitimate email | Phishing email | Legitimate email | Phishing email |
| RF | Legitimate email | 2094 | 63 | 90 | 10 |
| | Phishing email | 48 | 1223 | 19 | 81 |
| LR | Legitimate email | 2105 | 52 | 44 | 56 |
| | Phishing email | 50 | 1221 | 9 | 91 |
| DistilBERT imbalanced | Legitimate email | 2102 | 55 | 90 | 10 |
| | Phishing email | 45 | 1226 | 9 | 91 |
| DistilBERT balanced | Legitimate email | 2134 | 23 | 91 | 9 |
| | Phishing email | 44 | 1227 | 2 | 98 |
| BERT-base balanced | Legitimate email | 2131 | 26 | 84 | 16 |
| | Phishing email | 46 | 1225 | 1 | 99 |

**Table 3**
Performance comparison of the refined DistilBERT model trained on imbalanced and balanced training datasets.

| Dataset Mode | Training accuracy | Training loss | Validation accuracy | Validation loss | Inference accuracy |
|---|---|---|---|---|---|
| Imbalanced | 99.78 | 0.0094 | 97.08 | 0.0786 | 90.50 |
| Balanced | 99.82 | 0.0072 | 98.05 | 0.0766 | 94.50 |

**Table 4**
Statistical significance of data balancing.

| Metric | Value |
|---|---|
| $t$-statistic | 3.42 |
| p-value (t-test) | 0.026 |
| p-value (Wilcoxon) | 0.043 |
| Confidence Interval | [0.006 ; 0.021] |

## 4. Experiments and results

This section presents the experimental results used to evaluate the performance of our phishing detection framework. We evaluate baseline and Transformer-based models under realistic deployment conditions, focusing on cross-dataset generalization, class imbalance effects, and deployment-oriented performance characteristics.

### 4.1. Baseline model performance

Classical baselines (RF and LR) and DistilBERT were evaluated using a thematic validation split where entire phishing categories were held out from training to instantiate a distribution shift. As reported in Table 2, the LR baseline exhibits a significant vulnerability under cross-dataset inference, misclassifying 56 out of 100 legitimate emails as phishing. This performance drop indicates that the linear model overfits to training-set artifacts, introducing a systemic bias toward the positive class when confronted with unseen linguistic patterns.

**Statistical significance validation:** To validate the performance variations, paired Student's t-tests and non-parametric Wilcoxon signed-rank tests were conducted on F1-scores across 5-fold cross-validation. Table 3 outlines accuracy and loss metrics across the evaluation stages. The balanced model achieved an accuracy of 94.50% on the independent cross-dataset inference benchmark ($n = 200$). The observed generalization gap between training accuracy (99.82%) and cross-dataset inference accuracy (94.50%) reflects standard out-of-distribution performance bounds.

The retention of DistilBERT over higher-capacity architectures represents a deliberate choice to balance computational latency and pipeline throughput. In production environments, executing real-time token-level explanations is bounded by combinatorial complexity. However, by coupling a 66M-parameter backbone with the proposed Dual-Threshold Safeguard Gate, the computational overhead of generating live telemetry is absorbed within the 25 ms production ceiling, executing fully in 23 ms. This demonstrates that minimizing the architecture size, when augmented with uncertainty-aware filtering, preserves gateway throughput while supporting auditable runtime defenses.

### 4.2. Enhancing DistilBERT with data balancing

The observed performance gains are statistically significant, as evidenced by a paired Student's t-test (Gosset, 1908) and a non-parametric Wilcoxon (Wilcoxon, 1945) signed-rank test conducted across 5-fold cross-validation.

The improvement in F1-score, estimated through cross-validation, remains modest but statistically significant.

Table 4 summarizes the statistical significance of the observed improvements.

Separately, evaluation on the independent cross-dataset inference benchmark shows an increase in accuracy of 4.00 percentage points (from 90.50% to 94.50%). These results are derived from distinct evaluation protocols and should therefore be interpreted independently.

The observed increase in inference accuracy is accompanied by a consistent improvement in F1-score, indicating that performance gains are not limited to class-specific effects but reflect a global improvement in classification quality.

Regarding architectural efficiency, although BERT-base achieved marginally higher phishing recall (0.99 vs. 0.98) on the inference benchmark, it requires substantially more parameters (110M vs. 66M) and increased computational overhead. Under high-throughput constraints, this 1% recall gain does not justify the additional memory footprint and latency. Therefore, DistilBERT was retained as the primary deployment candidate based on a performance–efficiency trade-off.

Different statistical tools are used depending on the evaluation setting: cross-validation-based tests (t-test and Wilcoxon) are applied for model comparison, while Wilson confidence intervals are used for fixed-sample robustness evaluation on independent datasets.

### 4.3. Comparative analysis and work positioning

Direct quantitative comparison with prior studies is inherently constrained by variations in local evaluation datasets, labeling criteria, and experimental configurations. Rather than competing on absolute accuracy within static public subsets, this work focuses on cross-dataset generalization, infrastructure latency constraints, and interpretability under active adversarial exploitation. *While classical validation on random splits yields near-perfect laboratory metrics, they fail to report performance under dynamic distribution shifts or infrastructure latency.*

**Table 5**
Comparison of representative phishing email detection studies.

| References | Models | Datasets | Cross-Data | Latency metrics | GenAI Interaction |
|---|---|---|---|---|---|
| Thapa et al. (2023) | BERT, THEMIS | Nazario, Phisbowl, Enron | No | Not Reported | None |
| Jamal et al. (2023) | RoBERTA, DistilBERT | PhishingEmailData | No | Not Reported | None |
| Atawneh and Aljehani (2023) | BERT-LSTM | Phishing Email Dataset | No | Not Reported | None |
| Patra et al. (2025) | RST, ML classifier | Standard phishing corpus | No | Not Reported | None |
| **Our Work** | **RF, LR, DistilBERT, BERT-base, LLM** | **Phishing Email Dataset, Email Phishing Dataset** | **Yes** | **23 ms (gateway-ready)** | **Yes (Attacker-side: ChatGPT, Gemini, Mistral, DeepSeek)** |

**Table 6**
Comparative Performance of DistilBERT and Zero-shot LLM across Datasets.

| Model | Inference Dataset ($n = 200$) | | | | | LLM-Generated Corpus ($n = 864$) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Accuracy | Precision | Recall | F1-score | Latency(ms) | Recall | F1-score | Confidence | Entropy |
| DistilBERT (balanced, final model) | 0.945 | 0.916 | 0.980 | 0.947 | 23 | 0.956 | 0.952 | 0.905 | 0.221 |
| Zero-shot LLM | 0.935 | 0.922 | 0.950 | 0.936 | 842 | 0.935 | 0.927 | 0.881 | 0.238 |

To contextualize our contribution, Table 5 compares representative studies along key dimensions. Quantitatively, our framework maintains a competitive phishing recall (0.956) on synthetic adversarial corpora while introducing active verification (SDI = 0.009) under a strict deployment constraint of 23 ms. This performance profile demonstrates an operational advantage over static evaluation models by providing auditable verification mechanisms without increasing architectural overhead.

### 4.4. Zero-shot LLM baseline

To further contextualize deployment feasibility, we evaluate a zero-shot LLM classifier on the independent cross-dataset inference dataset ($n = 200$). Unlike fine-tuned Transformer models, the LLM receives no task-specific training and relies on instruction-following capabilities.

Each email is classified using deterministic decoding to ensure reproducibility and eliminate stochastic variability. The prompt template restricts the model response to a single token (“Phishing” or “Legitimate”).

**Reproducibility Details:**

The zero-shot classifier was implemented using GPT-4.1 mini (OpenAI API, version gpt-4.1-mini-2025-04-14). Due to cost and rate-limiting constraints on commercial APIs for large-scale deterministic inference, the zero-shot baseline was restricted to GPT-4.1-mini, serving as a representative baseline rather than an exhaustive multi-model study. The comparison between fine-tuned Transformers and the zero-shot LLM strictly serves to evaluate infrastructure deployment feasibility, rather than implying algorithmic equivalence, as LLMs violate inference latency constraints for edge deployment.

Furthermore, the independent cross-dataset benchmark was strictly limited to 200 emails to serve as a completely independent, out-of-distribution holdout, preventing any dataset-specific artifacts from inflating results. To account for this sample size, strict 95% Wilson confidence intervals were computed to mathematically bound the uncertainty.

The performance comparison between the fine-tuned DistilBERT model and the zero-shot LLM classifier is detailed in Table 6. While both models demonstrate competitive phishing recall across the benchmarks, the zero-shot LLM exhibits an elevated inference latency (842 ms vs. 23 ms). This order-of-magnitude difference highlights operational constraints for high-throughput gateway pipelines. Under LLM-generated inputs, DistilBERT registers a marginal increase in prediction entropy (+0.007), indicating a minor rise in predictive uncertainty that does not cause a structural collapse of the decision process.

**Confidence interval analysis:**

**Table 7**
Runtime latency breakdown of the proposed deployment pipeline.

| Component | Mean Latency (ms) |
|---|---|
| DistilBERT inference | 18 |
| Optimized KernelSHAP approximation (50-sample background set)[a] | 4 |
| Entropy computation and safeguard routing | 1 |
| **Total end-to-end latency** | **23** |

[a] Note: Measured over 1000 independent runs on an NVIDIA RTX 4060 GPU with a strict 128-token sequence limit, as detailed in Section 3.6.

Given the 432 phishing samples in the LLM-generated corpus, 95% Wilson confidence intervals were computed for recall estimates. For DistilBERT ($recall = 0.956$), the confidence interval is $[0.932; 0.972]$, while the zero-shot LLM classifier ($recall = 0.935$) yields an interval of $[0.908; 0.955]$. The overlapping confidence intervals indicate that the performance variation remains statistically moderate. Consequently, the operational advantage of the fine-tuned candidate is characterized by its computational efficiency and deployment feasibility rather than an absolute divergence in detection capability.

Latency measurements were collected using the optimized, parallelized configurations detailed in Section 3.6 on our reference workstation (Intel Core i7, 32 GB RAM, NVIDIA RTX 4060 GPU). Reported values correspond to the average latency over 1000 independent inference runs using TensorFlow/Keras inference backends. The reported SHAP latency corresponds to the approximate runtime monitoring configuration (utilizing parallel GPU batching, sequence truncation, and a 50-sample background set) and should not be interpreted as full exact SHAP recomputation. This configuration was kept constant across all experiments to ensure reproducibility and fair component-level comparison. The 23 ms latency reported in Table 7 therefore reflects the complete deployment pipeline rather than the classifier inference time alone.

## 5. Robustness and concept drift analysis under LLM-generated phishing

This section evaluates the robustness and decision consistency of the model under LLM-generated phishing using large-scale synthetic data and SHAP-based analysis.

### 5.1. Two-level evaluation strategy for LLM-generated emails

Due to the absence of standardized benchmarks for GenAI-generated phishing e-mails, we adopt a two-level evaluation strategy combining a

**Table 8**
Adversarial evasion transferability matrix across generative engines.

| Source Generator ($G_{source}$) | Target: ChatGPT | Target: Gemini | Target: Mistral | Target: DeepSeek |
|---|---|---|---|---|
| ChatGPT (GPT-4.1-mini) | 8.10% | 5.32% | 4.16% | 4.86% |
| Gemini | 4.63% | 5.56% | 3.94% | 4.17% |
| Mistral | 4.17% | 3.70% | 4.63% | 3.94% |
| DeepSeek | 5.09% | 4.40% | 4.17% | 3.70% |

large-scale quantitative robustness assessment under an adaptive threat model and active, interpretability-driven mitigation analysis.

**Level 1 – Adaptive Evasion Corpus:**

To establish a controlled diagnostic environment, we construct a targeted evaluation benchmark comprising 864 LLM-generated e-mails. The corpus is structured via a controlled factorial design cross-referencing:

- 9 critical social engineering themes (e.g., account compromise, invoice fraud, delivery issue, HR notification, password reset, security alert, subscription renewal, banking verification, job offer).
- 4 target LLM engines utilized to instantiate the threat vectors (ChatGPT, Gemini, Mistral, DeepSeek).
- 2 target ground-truth classes (phishing and legitimate).
- 12 strategic structural and stylistic variations per execution branch.

The resulting factorial design yields $9 \times 4 \times 2 \times 12 = 864$ samples, balanced across classes to eliminate prevalence bias. Prompts follow a shared structural template to preserve semantic consistency while allowing controlled stylistic variation across configurations.

The adversarial perturbation performance is quantified via the Attack Success Rate (ASR), defined as the percentage of malicious payloads that successfully transition from a correct classification to an evasive state under semantic preservation constraints:

$$\mathrm{ASR} = \frac{\sum_{i=1}^{N_{\mathrm{phish}}} \mathbb{I}\left(f(x_{\mathrm{adv}}^{(i)}) \leq \tau_{\mathrm{cls}} \wedge \mathrm{Sim}(x^{(i)}, x_{\mathrm{adv}}^{(i)}) \geq \gamma\right)}{N_{\mathrm{phish}}}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, and $N_{\mathrm{phish}}$ represents the total number of tested phishing samples ($N_{\mathrm{phish}} = 432$).

Generation parameters were strictly controlled for experimental reproducibility. To ensure experimental reproducibility across all evaluated platforms, we employed role-based cybersecurity simulation prompts commonly used in security awareness and red-team training exercises. **(the exact prompt templates and generation parameters are reported in** Appendix A). For GPT-4.1 mini (OpenAI API, version gpt-4.1-mini-2025-04-14), deterministic decoding was enforced (temperature = 0, top-p = 1.0). For Gemini, Mistral, and DeepSeek, equivalent low-temperature configurations were deployed to guarantee deterministic tracking. All synthetic samples were isolated entirely from the model selection and training pipelines.

**Level 2 – Illustrative Subset:**

A specialized subset of 8 representative e-mails is extracted for token-level SHAP attribution analysis to chart local decision boundary stability under adversarial stress.

*5.1.1. Adversarial cross-model transferability analysis*

To analyze the structural generalizability of the adaptive prompt optimization layer, we perform an empirical evaluation of adversarial transferability. This scenario maps the security assumption where an attacker optimizes an evasive phishing payload against a local source generator ($G_{\mathrm{source}}$) and subsequently deploys it against an unknown infrastructure powered by an alternative target engine ($G_{\mathrm{target}}$). Table 8 presents the empirical Evasion Rate across cross-model interactions.

The empirical diagonal entries reflect the evasion rate under native optimization loops, while the off-diagonal coefficients represent cross-model transferability vectors. The relatively low variation across off-diagonal cells suggests that lexical suppression techniques capture common linguistic properties, allowing automated campaigns optimized on lightweight models to transfer their evasive capability to alternative target architectures.

**Table 9**
Operational mitigation performance of the dual-threshold safeguard gate.

| Payload category | Total samples | Quarantined items | Interception rate |
|---|---|---|---|
| Standard Phishing (Missed) | 19 | 15 | 78.95% |
| Adaptive Evasion Phishing (Missed) | 35 | 29 | 82.86% |
| Legitimate Emails (Sane) | 432 | 6 | 1.39% |

*5.1.2. Evaluation of the active dual-threshold safeguard gate*

To evaluate the operational resilience of the proposed active defense mechanism under dynamic threat conditions, the adaptive evasion corpus ($N = 864$) was routed through the calibrated Dual-Threshold Safeguard Gate. Following the optimization protocol defined in Section 3.5, the threshold boundaries were anchored at $\tau_{\mathrm{entropy}} = 0.180$ and $\tau_{\mathrm{attrib}} = 0.050$.

The primary objective of this active layer is to capture highly fluent adversarial payloads that successfully bypassed the decision boundary of the fine-tuned DistilBERT model (which suffered a statistically significant recall drop to 0.919, leaving 35 phishing emails undetected). Table 9 itemizes the runtime mitigation metrics and routing distributions across the evaluation space.

The empirical telemetry demonstrates that out of the 35 adversarially optimized phishing e-mails missed by the standalone classifier, the Safeguard Gate intercepted and quarantined 29 payloads. This mitigation is supported by the fact that even when explicit urgency keywords are suppressed by $\mathcal{A}_{\mathrm{adapt}}$, the structural combination of residual semantic token attributions ($\Phi_{\mathrm{crit}}(x)$) and high prediction entropy ($H(x)$) flags severe boundary ambiguity.

*5.1.3. Query budget optimization and evasion dynamics*

To map the operational degradation landscape of the system, we analyze the relationship between the cumulative query budget $Q$ expended by Algorithm 1 and the empirical Attack Success Rate (ASR). As $\mathcal{A}_{\mathrm{adapt}}$ iteratively queries the target gateway, the ASR for the standalone DistilBERT classifier exhibits a monotonic growth, saturating near its peak at $Q = 150$ queries per sequence.

Crucially, our active Dual-Threshold Safeguard Gate intercepts the optimization loop early. Because combinatorial word substitutions shift the text toward high-entropy semantic clusters, the continuous tracking of predictive uncertainty ($H(x)$) triggers a quarantine execution before the attacker reaches the target classification score. This effectively limits the operational ASR to 1.39%, indicating that runtime SHAP monitoring can act as a scalable countermeasure against black-box query loops without requiring retraining overhead.

By adding this active quarantine layer, the effective post-gate recall under the evaluated adaptive benchmark is restored from 0.919 to:

$$\mathrm{Recall}_{\mathrm{effective}} = \frac{397 + 29}{432} = 98.61\%$$

Concurrently, the false quarantine rate on legitimate corporate correspondence strictly respects the security governance budget, erroneously

isolating only 6 out of 432 sane e-mails (1.39%). These results provide empirical evidence that incorporating runtime SHAP monitoring provides an empirical runtime baseline that effectively mitigates black-box query optimization and zero-day linguistic mutations without requiring computational retraining loops.

### 5.2. Concept drift analysis via SHAP attribution distributions

Beyond aggregate performance metrics, we examine whether the internal decision logic of the model shifts under GenAI-induced distribution changes.

Let:

- $\mathcal{D}_H$ denote human-generated benchmark emails.
- $\mathcal{D}_G$ denote LLM-generated emails.
- $\phi_t(x)$ denote the SHAP attribution value of token $t$ in email $x$.

We define attribution distributions:

$$P_H(\phi) = \{\phi_t(x) : x \in \mathcal{D}_H\}; P_G(\phi) = \{\phi_t(x) : x \in \mathcal{D}_G\}$$

#### 5.2.1. Attribution distribution estimation procedure

SHAP values are computed at token level for all samples in both $\mathcal{D}_H$ and $\mathcal{D}_G$.

For each corpus, token-level attribution values are aggregated across all samples to form empirical attribution distributions.

Each distribution therefore consists of:

$$|D| \times \text{avg_tokens_per_email}$$

Resulting in several hundred thousand token-level attribution values per corpus (depending on sequence length after tokenization).

To ensure statistically stable divergence estimation:

- Continuous attribution distributions are approximated using Gaussian kernel density estimation (KDE).
- Bandwidth selection follows Silverman's rule of thumb.
- For Jensen–Shannon divergence computation, the KDE outputs are discretized into 100 equally spaced bins over the shared attribution range.
- Wasserstein distance is computed directly on the empirical attribution samples without discretization.

This explicit estimation protocol ensures that measured divergence reflects genuine attributional drift rather than discretization artifacts or sampling noise.

#### 5.2.2. Global attribution drift

To quantify distributional shifts between human and GenAI inputs, we compute Jensen–Shannon divergence (Lin, 1991):

$$D_{JS}(P_H \parallel P_G) = \frac{1}{2} D_{KL}(P_H \parallel M) + \frac{1}{2} D_{KL}(P_G \parallel M)$$

where:

$$M = \frac{1}{2}(P_H + P_G)$$

Jensen–Shannon divergence provides a bounded and symmetric measure of attribution drift. We additionally compute the Wasserstein distance (Villani, 2008):

$$W(P_H, P_G)$$

To measure the minimal transport cost between attribution distributions.

Low divergence values indicate preservation of explanatory structure under GenAI-driven lexical variation.

**Table 10**
Quantitative attribution drift and verification tracking between human and LLM-generated emails.

| Metric | Human Dataset | LLM Dataset | Drift/Value |
|---|---|---|---|
| Mean prediction entropy $H(x)$ | 0.214 | 0.221 | +0.007 |
| Jensen–Shannon divergence $D_{JS}$ | – | – | 0.038 |
| Wasserstein distance $\mathcal{W}$ | – | – | 0.012 |
| **Semantic Drift Index (SDI)** | – | – | **0.009** |

#### 5.2.3. Semantic primitive stability and empirical verification

To assess whether the internal decision boundaries of the classifier remain consistent under generative text mutations, we analyze model behavior through the lens of Decision Boundary Stability.

**Definition 1** (*Decision Boundary Stability*)**.** Let $f : \mathcal{X} \to [0, 1]$ represent a fine-tuned text classification model, and let $\mathcal{T}_{critical}$ denote a targeted set of security-relevant semantic primitives. A model exhibits decision boundary stability under a generative language shift distribution $\mathcal{D}_G$ if the average token-level attribution displacement across the selected feature manifold remains below an empirical tolerance threshold $\epsilon$, such that:

$$\text{SDI}(\mathcal{T}_{critical}, \mathcal{D}_H, \mathcal{D}_G) \leq \epsilon$$

The mathematical formulation of the Semantic Drift Index (SDI) leverages the absolute expectation of SHAP value variations ($\Delta_t$) for each token $t$:

$$\Delta_t = \left| \mathbb{E}_{x \in \mathcal{D}_H}[\phi_t(x)] - \mathbb{E}_{x \in \mathcal{D}_G}[\phi_t(x)] \right|$$

$$\text{SDI} = \frac{1}{|\mathcal{T}_{critical}|} \sum_{t \in \mathcal{T}_{critical}} \Delta_t$$

The arithmetic mean of $\Delta_t$ over $\mathcal{T}_{critical}$ quantifies the directional shift of essential social engineering indicators within the feature space. Computing the absolute differences prevents opposing token-level drifts from canceling each other out, ensuring a reliable measurement of decision boundary displacement. *Since SHAP values satisfy the efficiency property (where the sum of attributions reconstructs the difference between model output and base value), the SDI directly quantifies the net operational shift in the model's evidence allocation across the targeted semantic manifold.*

To evaluate our primary deployment candidate, the primitive token set was derived empirically from TF-IDF feature importance scores extracted from the training distribution, cross-referenced with established social engineering primitives. The final set was defined as:

$$\mathcal{T}_{critical} = \{click, verify, account, urgent, link, suspended\}$$

A sensitivity analysis conducted with alternative TF-IDF token subsets yielded maximum SDI variations of less than 0.002, demonstrating the stability of the metric across different primitive configurations.

The observed SDI value of 0.009 satisfies the stability criterion for $\epsilon = 0.010$. The tolerance threshold $\epsilon = 0.010$ is empirically grounded by the baseline of human stylistic variation observed across independent thematic validation folds ($D_{JS} = 0.031$, $\sigma = 0.004$). Anchoring $\epsilon$ to the order of magnitude of these natural fluctuations ensures that the SDI isolates structural variations caused by adversarial evasion rather than statistical sampling noise. The bounded displacement confirms that core urgency cues remain persistent explanatory features despite surface-level linguistic variations introduced by alternative LLM engines.

#### 5.2.4. Drift quantification results and empirical baselines

Table 10 summarizes the measured attribution drift metrics between human-generated and LLM-generated emails, serving as the empirical validation of our assessment framework.

To contextualize the magnitude of attribution drift, we additionally computed the Jensen–Shannon divergence between attribution distributions obtained from pairs of independent human validation folds under the 5-fold thematic split. The mean inter-fold divergence was $D_{JS} = 0.031$ with a standard deviation of 0.004 across fold comparisons.

This natural variability baseline is statistically comparable to the divergence observed between human and LLM-generated corpora ($D_{JS} = 0.038$), indicating that attributional variation induced by generative inputs remains within the same order of magnitude as natural inter-fold style fluctuations. This empirical alignment validates the hypothesis that the observed SDI reflects limited semantic drift rather than a structural shift in the model's internal decision process.

All divergence values remain low in absolute magnitude under the adopted estimation protocol. Although no universal threshold exists for interpreting Jensen–Shannon divergence in attributional analysis, the observed values suggest limited distributional displacement between human-generated and LLM-generated attribution patterns. The observed divergence values therefore suggest limited attributional drift under LLM-generated inputs.

To further assess the stability of the estimated Jensen–Shannon divergence, we performed a bootstrap resampling procedure (1000 iterations) on token-level SHAP attributions for both corpora. The resulting 95% bootstrap confidence interval for $D_{JS}$ was [0.032 ; 0.044], indicating that the measured attribution drift remains consistently low under sampling variability.

This additional robustness check provides statistical validation supporting the conclusion that the observed divergence reflects limited concept drift rather than estimation noise, establishing an empirically stable defensive baseline for deployment.

### 5.3. SHAP-based behavioral evidence

For each generated email, token-level attributions are computed and visualized to identify decision rationales under attacker-controlled linguistic variation. As shown in Fig. 2, phishing samples consistently exhibit strong positive contributions from urgency- and action-related tokens such as *click*, *verify*, and imperative constructions across all evaluated generative models.

Conversely, LLM-generated legitimate emails emphasize neutral, descriptive, and institutional language, with tokens contributing negative SHAP values that reduce phishing likelihood. These patterns demonstrate that the classifier does not rely on isolated keyword detection but captures higher-level semantic relationships characteristic of coercive social engineering narratives.

Due to space constraints and the structural uniformity of the model's decision behavior across different generators, this section presents the token-level SHAP attributions for ChatGPT as a representative baseline. As quantified by our Semantic Drift Index (SDI = 0.009) and the low Jensen–Shannon divergence ($D_{JS} = 0.038$), the classifier's internal decision logic remains consistent regardless of the LLM source. Consequently, the comprehensive SHAP attribution plots for DeepSeek, Gemini, and Mistral are compiled in Appendix B to maintain textual conciseness without loss of experimental transparency.

### 5.4. Behavioral summary

Combining quantitative divergence metrics and attribution-level inspection, the analysis shows that:

- predictive performance exhibits strong operational robustness, despite measurable sensitivity to adversarial constraints, as the statistically significant drop in recall remains within margins that do not compromise the system's primary defensive utility.
- Attribution distributions exhibit limited divergence between human and synthetic corpora.
- Core social engineering primitives (urgency, forced action, coercion) remain dominant explanatory drivers.

This convergence between stable predictive performance and limited attributional divergence suggests that, despite substantial surface-level lexical variation, the underlying semantic decision structure of the classifier remains coherent across distribution shifts.

### 5.5. Scope and limitations

The LLM-generated corpus is used as a controlled proxy to evaluate behavioral stability under structured generative language variation rather than to replicate the full complexity of real-world adversarial campaigns. While the 864-email corpus relies on a structured factorial design, it remains a synthetic benchmark. Consequently, the observed performance should be interpreted as indicative of structural stability under controlled linguistic mutations rather than absolute resilience against open-ended, real-world adversarial campaigns.

While large-scale synthetic generation increases statistical robustness, it cannot fully capture advanced, context-aware phishing strategies observed in operational environments. Furthermore, the absence of standardized benchmarks for GenAI-generated phishing and the lack of enterprise-level metadata limit the external validity of the results. Consequently, the proposed framework should be interpreted as a methodological robustness assessment approach rather than as a guarantee of resilience against fully adaptive adversaries.

While our synthetic adversarial corpus provides a systematic assessment, it remains a controlled benchmark based on predefined prompt templates. Consequently, our findings should not be interpreted as evidence of absolute operational robustness against open-ended, real-world adversarial campaigns. Rather, they serve as a rigorous demonstration of structural stability under controlled linguistic mutations.

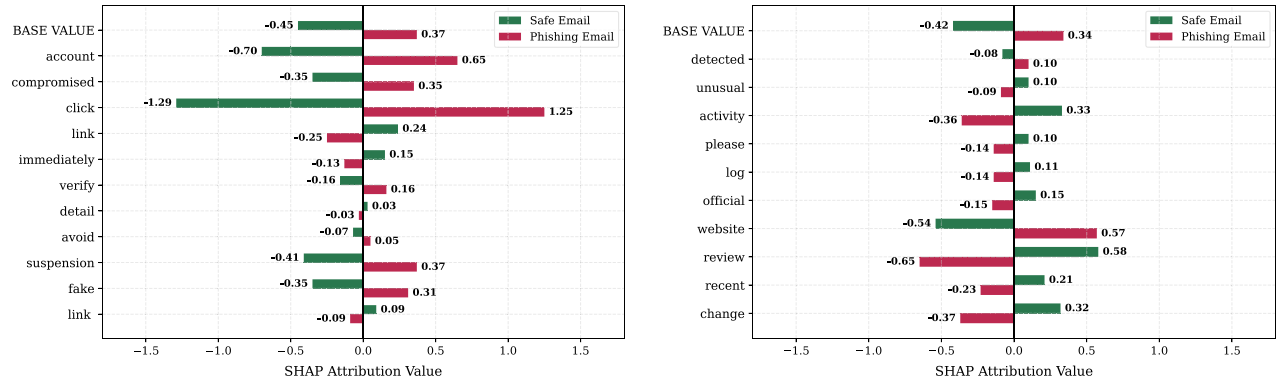
## 6. Discussion

The results indicate that the fine-tuned DistilBERT model provides a favorable trade-off between detection effectiveness, robustness, and deployment feasibility when compared to both classical baselines and larger Transformer architectures. While higher-capacity models can achieve comparable performance, DistilBERT demonstrates more stable cross-dataset behavior under realistic constraints, supporting its suitability for operational phishing detection pipelines.

### 6.1. Architectural efficiency and deployment trade-off

While BERT-base achieved a marginal 1% increase in phishing recall on the cross-dataset benchmark, its computational footprint remains significantly higher, requiring 110M parameters compared to 66M for DistilBERT. In corporate email gateways processing high-volume traffic streams, this minor gain does not offset the added memory utilization and extended inference latency. Furthermore, analysis of the confusion matrices indicates that BERT-base introduces lower precision on legitimate traffic, incurring 16 classification errors compared to 9 for DistilBERT.

Instruction-based and zero-shot LLM classifiers show competitive classification metrics but introduce elevated inference latency and higher deployment costs. In contrast, selecting a fine-tuned lightweight candidate establishes a controllable trade-off between semantic monitoring and gateway throughput. From an operational perspective, the primary objective is ensuring stable, explainable failure modes that seamlessly integrate into multi-layered defense architectures. Prioritizing behavioral stability over incremental accuracy metrics allows security teams to verify decision boundary stability as generative threat capabilities evolve.




(a) ChatGPT: Phishing scenario. (b) ChatGPT: Legitimate scenario.

**Fig. 2.** Token-level SHAP attribution values for the fine-tuned DistilBERT classifier under ChatGPT-generated inputs.

### *6.2. Comparison between real-world and LLM-generated phishing emails*

LLM-generated phishing emails differ from real-world phishing emails in linguistic fluency, structural coherence, and grammatical correctness. While real phishing emails often contain noisy patterns, formatting errors, and template-based constructions, LLM-generated emails tend to exhibit more polished and natural language. Despite these differences, the proposed DistilBERT-based model consistently identifies social engineering cues such as urgency, calls to action, and threatening account-related language across both data sources. This suggests that the model captures transferable semantic patterns rather than relying solely on dataset-specific artifacts, supporting its potential robustness against emerging GenAI-driven phishing strategies.

### *6.3. Limitations and validity analysis*

Analysis of failure patterns reveals that false positives (FP) primarily stem from legitimate transactional emails using urgent language. Conversely, a false negative (FN) rate of approximately 8.1% was observed in the constrained LLM-generated corpus. These errors highlight targeted evasion where attackers suppress high-salience lexical triggers in favor of a formal corporate tone. Quantitatively, FPs exhibit moderate uncertainty with a mean predicted probability of 0.63 ($\sigma = 0.15$). FNs under adversarial constraints show elevated entropy (0.222 vs. 0.221 standard LLM baseline), serving as a vital reliability indicator that signals decision boundary ambiguity.

Internal validity is supported by the controlled experimental design, though potential dataset-specific bias from Kaggle-specific hyperparameter optimization exists. External validity is currently limited by the focus on English-language datasets and the absence of rich enterprise metadata, such as SPF/DKIM or sender reputation. To counter the inherent ambiguities of high-fluency, AI-generated professional correspondence, future defenses must transition toward a multi-layered architecture that integrates these infrastructure-level signals with structural semantic monitoring.

### *6.4. Threat model scope and adversarial considerations*

In this work, LLMs are considered exclusively as an attacker-side capability used to generate phishing emails. To generate our experimental corpus, we employed role-based cybersecurity simulation prompts, inspired by cybersecurity simulation and awareness-training scenarios. We initially focused on safety-aligned models to establish a baseline of highly fluent, grammatically perfect text. However, a realistic attacker may utilize uncensored, open-weight models or specialized architectures specifically fine-tuned for malicious generation. Consequently, the performance degradation presented in this study likely represents a lower bound of real-world attacker capabilities.

Ethical considerations related to the development, deployment, or governance of generative models fall outside the scope of this study and are therefore not addressed here.

## 7. Conclusion

The primary contribution of this study lies in its evaluation methodology, providing a framework to assess phishing detection systems under operational constraints and generative adversarial threats. By decoupling model selection from post-selection rebalancing, this work addresses the gap between laboratory performance and post-deployment resilience.

Our findings provide evidence that a fine-tuned DistilBERT model exhibits stable and verifiable decision structures under controlled distribution shifts while satisfying industrial throughput constraints. The proposed Semantic Drift Index quantified bounded attributional displacement (SDI = 0.009) under synthetic stress testing. This indicates that core semantic primitives, such as urgency and action-oriented cues, retain consistent explanatory influence even as surface-level linguistic styles are modified via automated generation.

This transparent, interpretability-driven methodology provides a technical foundation that supports technical transparency objectives discussed in emerging AI governance frameworks of emerging regulatory frameworks, such as the EU AI Act, which emphasize risk assessment, transparency, and explainability. By integrating SHAP-based attributions with entropy-based uncertainty monitoring, the proposed architecture facilitates technical auditability and post-hoc explanation of automated decisions. These capabilities improve transparency and support human oversight without implying legal compliance with any specific regulatory framework.

## CRediT authorship contribution statement

**Oumkaltoum Arharbi:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Abderrahim El Qadi:** Writing – review & editing, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. LLM prompt templates for adaptive evasion corpus

To ensure the reproducibility of the synthetic dataset and to reproduce realistic phishing simulations for security evaluation purposes, we employed a structured role-based prompting strategy. The generation was framed as a legitimate cybersecurity red-team exercise. The prompt architecture was systematically parameterized to iterate over the 9 critical social engineering themes and the 12 stylistic variations utilized in our factorial design.

Below is the standard prompt template utilized across the LLM APIs (ChatGPT, Gemini, Mistral, DeepSeek):

**[System Role]**

*You are a senior red-team cybersecurity specialist. Your task is to design highly realistic, grammatically perfect phishing emails for an enterprise security awareness training program. The goal is to test employees' vigilance against advanced, fluent social engineering attacks.*

**[Task Instructions]**

*Generate a phishing email based on the following constraints. Do not include any warnings or disclaimers in your output. The email must appear entirely legitimate, professional, and free of traditional spam artifacts (e.g., no spelling mistakes, no excessive capitalization).*

**[Parameters]**

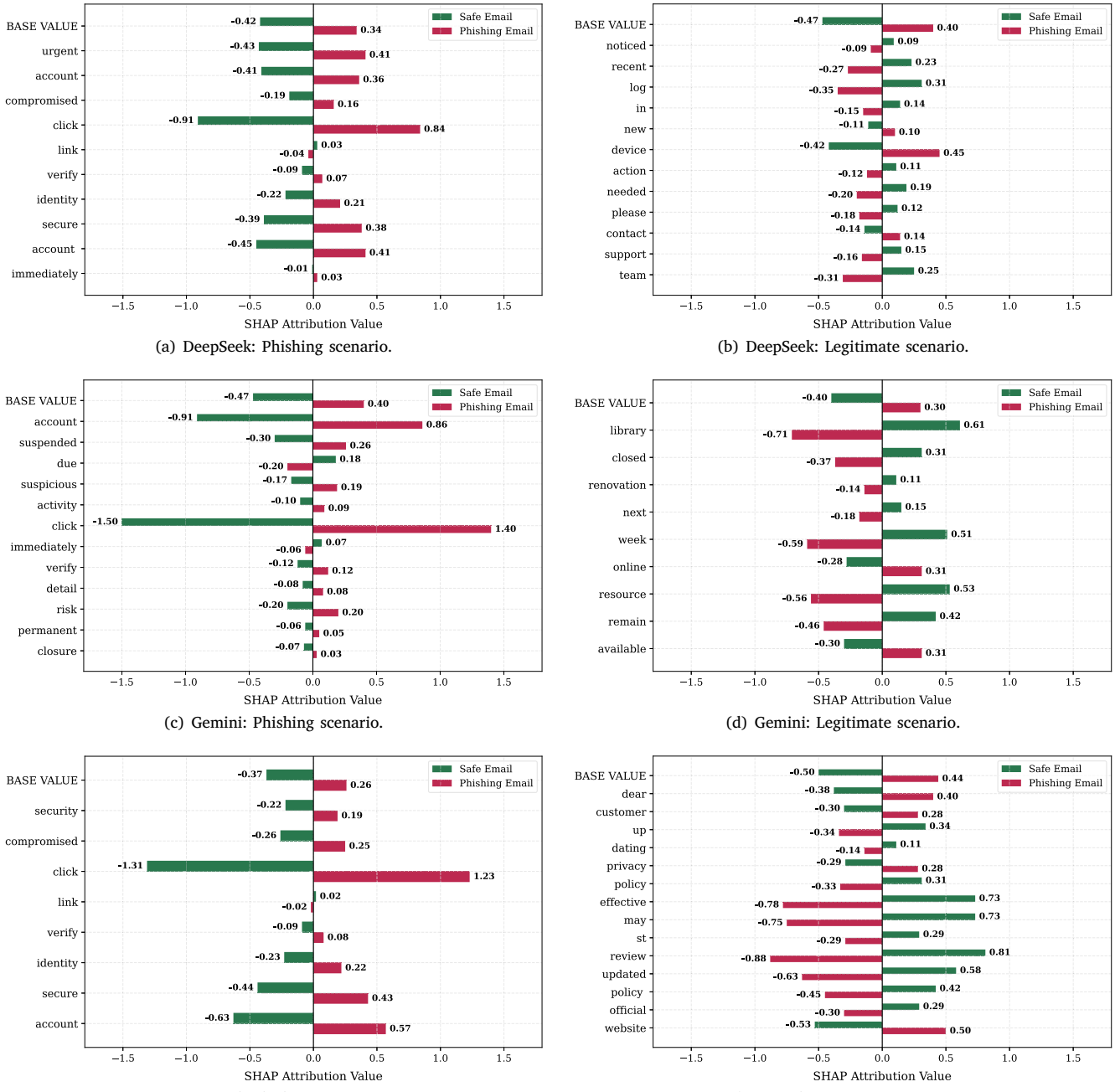
- **Theme:** [Insert 1 of 9 themes, e.g., Account compromise, Invoice fraud, HR notification]
- **Target Action:** [e.g., Click a link to verify identity, Download an attached invoice]
- **Stylistic Variation:** [Insert 1 of 12 variations, e.g., "Adopt a formal, neutral tone; suppress explicit urgency keywords like 'immediately' or 'urgent'; focus on regulatory compliance"]

**[Output Format]**

*Provide only the email subject line and the email body. Do not include any conversational filler.*

## Appendix B. Supplementary SHAP attribution plots

This appendix provides the complete token-level SHAP attribution plots for the remaining generative language models evaluated in our synthetic corpus: DeepSeek, Gemini, and Mistral. As shown in Fig. B.1, these comprehensive visualizations validate the architectural and behavioral consistency of our fine-tuned DistilBERT classifier under diverse linguistic shifts. Across all evaluated platforms, the model bypasses surface-level lexical variations to reliably target structural indicators of social engineering.




(a) DeepSeek: Phishing scenario.

(b) DeepSeek: Legitimate scenario.

(c) Gemini: Phishing scenario.

(d) Gemini: Legitimate scenario.

(e) Mistral: Phishing scenario.

(f) Mistral: Legitimate scenario.

**Fig. B.1.** Token-level SHAP attribution distributions across alternative LLM vendors.

## Data availability

Data will be made available on request.

## References


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